

$$Q2) \quad dl^2 = c^2 dt^2 - a^2 \left(\frac{d\sigma^2}{1-k\sigma^2} + \sigma^2 d\theta^2 + \sigma^2 \sin^2 \theta d\phi^2 \right)$$

for 3 sphere pick a time slice and set $k=1$. (1)

$$\Rightarrow ds^2 = a^2 \left(\frac{d\sigma^2}{1-\sigma^2} + \sigma^2 d\theta^2 + \sigma^2 \sin^2 \theta d\phi^2 \right)$$

$$Vol = \int \sqrt{\det(ds^2)} d\sigma d\theta d\phi \quad (1)$$

$$= \int \sqrt{\frac{a^2}{1-\sigma^2} \times a^2 \sigma^2 \times a^2 \sigma^2 \sin^2 \theta} d\sigma d\theta d\phi$$

$$= a^3 \int \frac{\sigma^2 \sin \theta}{\sqrt{1-\sigma^2}} d\sigma d\theta d\phi$$

$$= a^3 2\pi \times \int_0^1 \frac{\sigma^2 d\sigma}{\sqrt{1-\sigma^2}} \int_0^\pi \sin \theta d\theta \quad \sigma = \sin t$$

$$= 4\pi a^3 \int_0^{\pi/2} \frac{\sin^2 t \cancel{\cos t} dt}{\cancel{\cos t}}$$

$$\cos 2t = 1 - 2\sin^2 t$$

$$\Rightarrow \sin^2 t = \frac{1 + \cos 2t}{2}$$

$$= 4\pi a^3 \int_0^{\pi/2} \frac{1 + \cos 2t}{2} dt$$

$$= 4\pi a^3 \left[\frac{t}{1} + \frac{1}{2} \frac{\sin 2t}{2} \right]_0^{\pi/2}$$

$$= 4\pi a^3 \left[\frac{\pi}{4} + \frac{1}{4} (0 - 0) \right]$$

$$= \pi^2 a^3$$

(1)

Q1. $\frac{1}{r} = \frac{GM}{l^2} [1 + e \cos(\phi - e\phi)]$

$$e = \left(\frac{3GM}{lc} \right)^2$$

$$l = \sqrt{Gma(1-e^2)}$$

Neptune's orbit: $T_N = 88 \text{ day}$

$$e_N = 0.21$$

$$\Delta\phi_N = 43'' \text{ per } 100 \text{ years}$$

Pluto's orbit: $T_P = 248 \text{ yr}$

$$e_P = 0.25$$

$$\frac{\Delta\phi}{\text{per orbit}} = \frac{K}{a(1-e^2)}$$

in ΔT : $n_{\text{orb}} = \frac{\Delta T}{T}$

$$\Delta\phi_p \text{ in } \Delta T = \frac{k \Delta T}{T_p a_p (1 - e_p^2)}$$

$$\Delta\phi_M \text{ in } \Delta T = \frac{k \Delta T}{T_M a_M (1 - e_M^2)}$$

$$\frac{\Delta\phi_p}{\Delta\phi_M} = \frac{T_M a_M (1 - e_M^2)}{T_p a_p (1 - e_p^2)}$$

$$a^3 \propto T^2 \quad \rightarrow \quad \underline{\underline{a \propto T^{2/3}}}$$

$$\frac{\Delta\phi_p}{\Delta\phi_M} = \frac{T_M^{5/3} (1 - e_M^2)}{T_p^{5/3} (1 - e_p^2)}$$

$$\Delta\phi_p \text{ in } 100 \text{ yr} = \frac{T_M^{5/3} (1 - e_M^2)}{T_p^{5/3} (1 - e_p^2)} \times 43''$$

$$\underline{\underline{\Delta\phi_p \text{ in } 248 \text{ yr}}} = \frac{\overset{\text{red } T_p}{248}}{100} \times \frac{T_M^{5/3} (1 - e_M^2)}{T_p^{5/3} (1 - e_p^2)} \times 43''$$

$$\frac{\Delta\phi_p}{\text{in 248 yr}} = \frac{43''}{100} \times \frac{T_M^{5/3} (1 - e_M^2)}{T_P^{2/3} (1 - e_P^2)} - (*)$$

alternatively:

$$\frac{\Delta\phi_p}{\Delta\phi_M} = \frac{a_M (1 - e_M^2)}{a_P (1 - e_P^2)}$$

→ (1 pt.) for expression

(*) same expression as (**)

$$= \frac{T_M^{2/3} (1 - e_M^2)}{T_P^{2/3} (1 - e_P^2)}$$

$$\Delta\phi_p^0 = \frac{T_M^{2/3} (1 - e_M^2)}{T_P^{2/3} (1 - e_P^2)} \times \frac{\Delta\phi_M \times T_M}{100}$$

$$(**) \leftarrow = \frac{T_M^{5/3} (1 - e_M^2)}{T_P^{2/3} (1 - e_P^2)} \times \frac{43''}{100}$$

$$= 0.00103'' \text{ per orbit}$$

(1 pt.) for calculation.

(0.5 pt.) removed if units issue

2026

Quiz 2 - Q3

Metric around SMBH

$$ds^2 = \left[1 - \frac{R_s}{r}\right] dt^2 - \frac{dr^2}{\left[1 - \frac{R_s}{r}\right]} - r^2 d\Omega^2$$

Consider a circular orbit with $r = R$, $\theta = \pi/2$
 $c = 1$

$$\frac{d\phi}{dt} = \Omega$$

$$\textcircled{a} \quad u^\mu = \frac{dx^\mu}{d\tau} = [u^t, 0, 0, u^\phi]$$

$$u^\phi = \frac{d\phi}{d\tau} = \frac{d\phi}{dt} \frac{dt}{d\tau} = \Omega u^t$$

$$\textcircled{1} \quad \begin{aligned} u^\mu &= [u^t, 0, 0, \Omega u^t] & u_\mu &= \left[\left(1 - \frac{R_s}{R}\right)u^t, 0, 0, R\Omega u^t\right] \\ u^\mu u_\mu &= u^t u_t - u^\phi u_\phi = \left(1 - \frac{R_s}{R}\right)u^t{}^2 - \Omega^2 R^2 u^t{}^2 \\ &= \left(\left(1 - \frac{R_s}{R}\right) - \Omega^2 R^2\right)u^t{}^2 = 1 \end{aligned}$$

$$\therefore u^t = \frac{1}{\sqrt{\left(1 - \frac{R_s}{R}\right) - \Omega^2 R^2}}$$

$\textcircled{0.5} \rightarrow$ But Ω & R are fixed
 $\therefore u^t = \text{constant}$.

$$a^\mu = u^\nu \nabla_\nu u^\mu = u^\mu [\partial_\nu u^\mu + \Gamma_{\nu\lambda}^\mu u^\lambda]$$

But $u^t = \text{constant}$

$$\therefore \partial_\nu u^\mu = 0.$$

$$a^\mu = u^\nu \Gamma_{\nu\sigma}^\mu u^\sigma$$

$$= \Gamma_{\nu\sigma}^\mu u^\sigma u^\nu \rightarrow \textcircled{0.5}$$

If u^t & u^φ are only nonzero then
 Γ_{tt}^μ or $\Gamma_{\varphi\varphi}^\mu$ can't have t, φ as metric
 doesn't depend on φ &
 it is diagonal.

$$a^\mu = \Gamma_{tt}^\mu u^t{}^2 + \Gamma_{\varphi\varphi}^\mu u^\varphi{}^2$$

$\Rightarrow \mu = r$ as metric depends on r

$$a^r = \Gamma_{tt}^r u^t{}^2 + \Gamma_{\varphi\varphi}^r u^\varphi{}^2$$

$$\textcircled{1} \begin{cases} \Gamma_{tt}^r = \frac{1}{2} g^{rr} \left[-\frac{\partial g_{tt}}{\partial r} \right] = \frac{1}{2} \left(1 - \frac{R_s}{r} \right) \left[+\frac{R_s}{r^2} \right] \\ \Gamma_{\varphi\varphi}^r = \frac{1}{2} g^{rr} \left[-\frac{\partial g_{\varphi\varphi}}{\partial r} \right] = -\frac{1}{2} \left(1 - \frac{R_s}{r} \right) [2r] \end{cases}$$

$$\therefore a^r = \frac{1}{2} \left(1 - \frac{R_s}{r} \right) \left[\frac{R_s}{r^2} u^t{}^2 - 2R\Omega^2 u^t{}^2 \right] - \textcircled{0.5}$$

$$a^r = \left(1 - \frac{R_s}{R}\right) \left[\frac{R_s}{2R^2} - R\Omega^2 \right] u^{t^2} \quad \left. \vphantom{a^r} \right\} \textcircled{8.5}$$

$$\sqrt{a^\mu a_\mu} = \sqrt{g_{rr} a^{r^2}} = \sqrt{g_{rr}} a^r = a$$

⑥ For circular case

$$a^r = 0$$

$$\Rightarrow \frac{GM}{R^2} = R\Omega^2$$

$$\Rightarrow \boxed{\Omega^2 R^3 = GM} \rightarrow \textcircled{8.5}$$

Yes, it is freely falling orbit. $\rightarrow \textcircled{8.5}$
as $a^r = 0$.